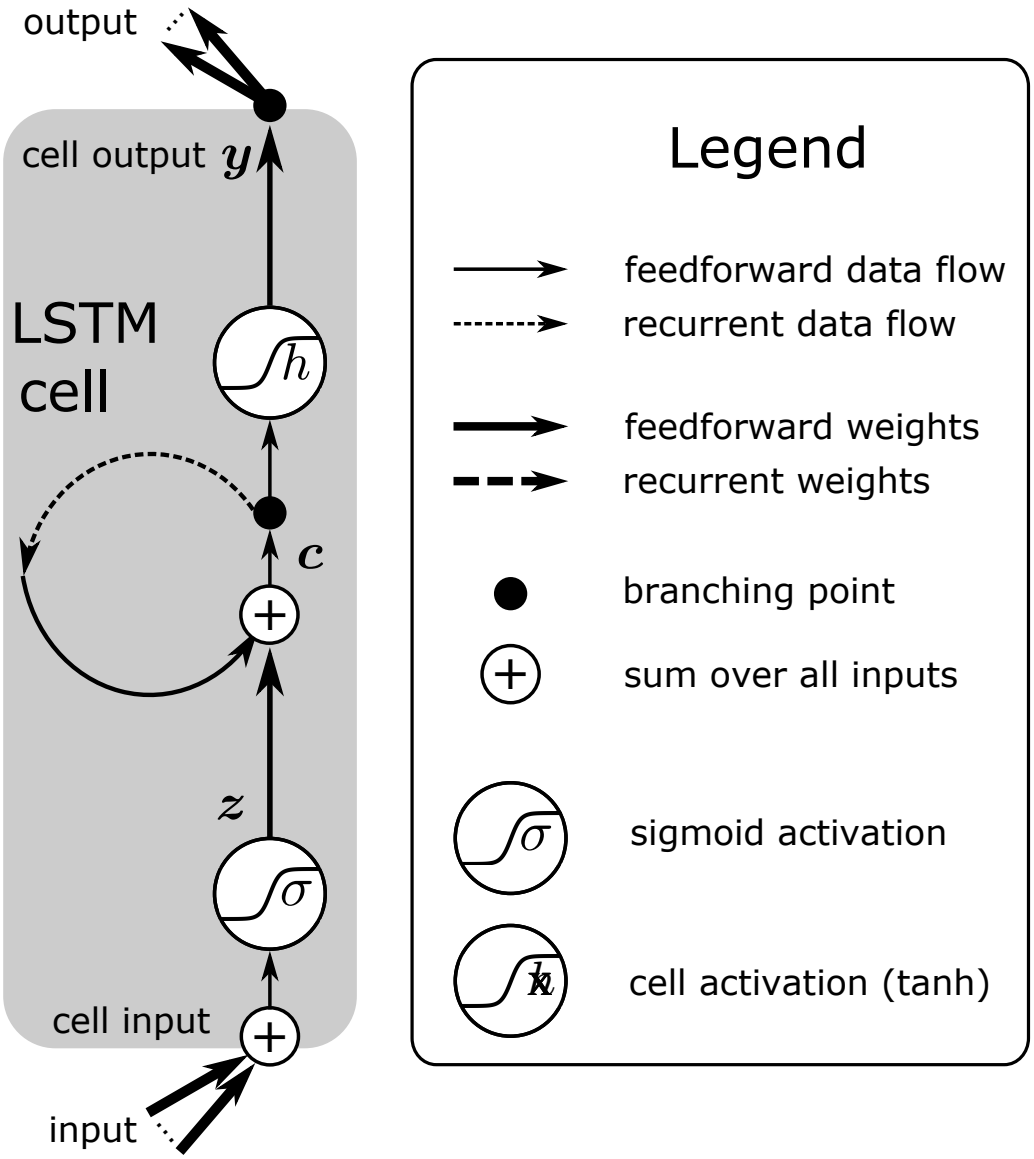

Figure A14: LSTM memory cell without gates.

A9 Contribution Analysis

A9.1 Difference of Consecutive Predictions for Sequences

General Approach. The idea is to assess the information gain that is induced by an input at a particular time step. This information gain is used for predicting the target at sequence end by determining the change in prediction. The input to a recurrent neural network is the sequence $\boldsymbol{x} = (x_1, \ldots, x_d)$ with target y_d, which is only given at sequence end. The prefix sequence $\boldsymbol{x}_t$ of length $t \leqslant d$ is $\boldsymbol{x}_t = (x_1, \ldots, x_t)$. F predicts the target y_d at every time step t:

$$F(\boldsymbol{x}_t) \;=\; y_d \;. \tag{A361}$$

We can define the decomposition of F through contributions at different time steps

$$h_0 \;=\; F(\boldsymbol{x}_0) \;, \tag{A362}$$

$$h_t \;=\; F(\boldsymbol{x}_t) \;-\; F(\boldsymbol{x}_{t-1}) \;\text{ for } t > 0 \;, \tag{A363}$$

where $F(\boldsymbol{x}_0)$ is a predefined constant. We have

$$F(\boldsymbol{x}_t) \;=\; \sum_{\tau=0}^{t} h_\tau \;. \tag{A364}$$

We assume a loss function for F that is minimal if $F \equiv F_{\min}$ predicts the expected y_d

$$F_{\min}(\boldsymbol{x}_t) \;=\; \mathrm{E}\left[y_d \mid \boldsymbol{x}_t\right] \;. \tag{A365}$$

Then

$$h_0 \;=\; \mathrm{E}\left[y_d\right] \;, \tag{A366}$$

$$h_t \;=\; \mathrm{E}\left[y_d \mid \boldsymbol{x}_t\right] \;-\; \mathrm{E}\left[y_d \mid \boldsymbol{x}_{t-1}\right] \;\text{ for } t > 0 \;. \tag{A367}$$

In this case, the contributions are the change in the expectation of the target that will be observed at sequence end. The contribution can be viewed as the information gain in time step t for predicting the target. If we cannot ensure that F predicts the target at every time step, then other contribution analysis methods must be employed. For attributing the prediction of a deep network to its input features several contribution analysis methods have been proposed. We consider Input Zeroing, Integrated Gradients (IG), and Layer-Wise Relevance Propagation (LRP).

Linear Models and Coefficient of Determination. We consider linear models and the average gain of information about the reward at sequence end if we go one time step further in the input sequence. By adding a variable, that is, another sequence element, the mean squared error (MSE) decreases, which is the amount by which the expectation improves due to new information. But by what amount does the MSE decrease in average? Here, we consider linear models. For linear models we are interested in how much the coefficient of determination increases if we add another variable, that is, if we see another input.
We consider the feature vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_k)^T$ from which the target y (the reward at sequence end) has to be predicted. We assume to have n pairs $(\boldsymbol{x}_i, y_i), 1 \leqslant i \leqslant n$, as training set. The prediction or estimation of y_i from $\boldsymbol{x}_i$ is $\hat{y}_i$ with $\hat{y}_i = F(\boldsymbol{x}_i)$. The vector of all training labels is $\boldsymbol{y} = (y_1, \ldots, y_n)$ and the training feature matrix is $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n)$. We define the mean squared error (MSE) as

$$\mathrm{mse}(\boldsymbol{y}, \boldsymbol{X}) \;=\; \frac{1}{n-1} \sum_{i=1}^{n} \left(\hat{y}_i \;-\; y_i\right)^2 \;. \tag{A368}$$

The *coefficient of determination* R^2 is equal to the correlation between the target y and its prediction $\hat{y}$. R^2 is given by:

$$R^2 \;=\; 1 \;-\; \frac{\frac{1}{n-1} \sum_{i=1}^{n} \left(\hat{y}_i \;-\; y_i\right)^2}{\frac{1}{n-1} \sum_{i=1}^{n} \left(y_i \;-\; \bar{y}\right)^2} \;=\; 1 \;-\; \frac{\mathrm{mse}(\boldsymbol{y}, \boldsymbol{X})}{s_y^2} \;. \tag{A369}$$

Therefore, R^2 is one minus the ratio of the mean squared error divided by the mean total sum of squares. R^2 is a strict monotonically decreasing function of the mean squared error.
We will give a breakdown of the factors that determine how much each variable adds to R^2 [100, chapter 10.6, p. 263]. The feature vector $\boldsymbol{x}$ is expanded by one additional feature z:

$\boldsymbol{w} = (x_1, x_2, \ldots, x_k, z)^T = (\boldsymbol{x}^T, z)^T$. We want to know the increase in R^2 due to adding z. Therefore, we decompose $\boldsymbol{w}$ into $\boldsymbol{x}$ and z. The difference in coefficients of determination is the difference of the according MSEs divided by the empirical variance of y:

$$R^2_{y\boldsymbol{w}} \; - \; R^2_{y\boldsymbol{x}} \; = \; \frac{\mathrm{mse}(\boldsymbol{y}, \boldsymbol{W}) \; - \; \mathrm{mse}(\boldsymbol{y}, \boldsymbol{X})}{s_y^2} \; . \tag{A370}$$

We further need definitions:

- $\boldsymbol{x} = (x_1, x_2, \ldots, x_k)^T$.
- $\boldsymbol{w} = (x_1, x_2, \ldots, x_k, z)^T = (\boldsymbol{x}^T, z)^T$.
- The sample covariance between y and x is $s_{yx} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})/(n-1)$, where $\bar{x} = \sum_{i=1}^n x_i/n$ and $\bar{y} = \sum_{i=1}^n y_i/n$ are the sample means. The variance of x is s_{xx} often written as s_x^2, the standard deviation squared: $s_x := \sqrt{s_{xx}}$.
- The correlation between y and x is $r_{yx} = s_{yx}/(s_x s_y)$.
- The covariance matrix $\boldsymbol{S}_{xx}$ of a vector $\boldsymbol{x}$ is the matrix with entries $[\boldsymbol{S}_{xx}]_{ij} = s_{x_i x_j}$.
- The covariance matrix $\boldsymbol{R}_{xx}$ of a vector $\boldsymbol{x}$ is the matrix with entries $[\boldsymbol{R}_{xx}]_{ij} = r_{x_i x_j}$.
- The diagonal matrix $\boldsymbol{D}_x = [\mathrm{diag}(\boldsymbol{S}_{xx})]^{1/2}$ has a ith diagonal entry $\sqrt{s_{x_i}}$ and is the diagonal matrix of standard deviations of the components of $\boldsymbol{x}$.
- R^2_{yw} is the squared multiple correlation between y and $\boldsymbol{w}$.
- R^2_{yx} is the squared multiple correlation between y and $\boldsymbol{x}$.
- $R^2_{zx} = \boldsymbol{s}_{zx}^T \boldsymbol{S}_{xx}^{-1} \boldsymbol{s}_{zx}/s_z^2 = \boldsymbol{r}_{zx}^T \boldsymbol{R}_{xx}^{-1} \boldsymbol{r}_{zx}$ is the squared multiple correlation between z and $\boldsymbol{x}$.
- r_{yz} is the simple correlation between y and z: $r_{yz} = s_{yz}/(s_y s_z)$.
- $\boldsymbol{r}_{yx} = (r_{yx_1}, r_{yx_2}, \ldots, r_{yx_k})^T = s_y^{-1} \boldsymbol{D}_x^{-1} \boldsymbol{S}_{yx}$ is the vector of correlations between y and $\boldsymbol{x}$.
- $\boldsymbol{r}_{zx} = (r_{zx_1}, r_{zx_2}, \ldots, r_{zx_k})^T = s_z^{-1} \boldsymbol{D}_x^{-1} \boldsymbol{S}_{zx}$ is the vector of correlations between z and $\boldsymbol{x}$.
- $\hat{\boldsymbol{\beta}}^*_{zx} = \boldsymbol{R}_{xx}^{-1} \boldsymbol{r}_{zx}$ is the vector of standardized regression coefficients (beta weights) of z regressed on $\boldsymbol{x}$.
- The parameter vector is partitioned into the constant β_0 and $\boldsymbol{\beta}_1$ via $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_m)^T = (\beta_0, \boldsymbol{\beta}_1^T)^T$. We have for the maximum likelihood estimate

$$\hat{\beta}_0 \; = \; \bar{y} \; - \; \boldsymbol{s}_{yx}^T \boldsymbol{S}_{xx}^{-1} \bar{\boldsymbol{x}} \; , \tag{A371}$$

$$\hat{\boldsymbol{\beta}}_1 \; = \; \boldsymbol{S}_{xx}^{-1} \boldsymbol{s}_{yx} \; . \tag{A372}$$

The offset $\hat{\beta}_0$ guarantees $\bar{\hat{y}} = \bar{y}$, therefore, $\boldsymbol{y}^T \bar{\boldsymbol{y}} = \hat{\boldsymbol{y}}^T \bar{\boldsymbol{y}}$, since $\bar{\boldsymbol{y}} = \bar{y}\mathbf{1}$:

$$\begin{aligned} \bar{\hat{y}} \; &= \; \frac{1}{n} \sum_{i=1}^n \hat{y}_i \; = \; \frac{1}{n} \sum_{i=1}^n \left(\hat{\beta}_0 \; + \; \hat{\boldsymbol{\beta}}_1^T \boldsymbol{x}_i \right) \\ &= \; \bar{y} \; - \; \boldsymbol{s}_{yx}^T \boldsymbol{S}_{xx}^{-1} \bar{\boldsymbol{x}} \; + \; \frac{1}{n} \sum_{i=1}^n \hat{\boldsymbol{\beta}}_1^T \boldsymbol{x}_i \\ &= \; \bar{y} \; - \; \boldsymbol{s}_{yx}^T \boldsymbol{S}_{xx}^{-1} \bar{\boldsymbol{x}} \; + \; \boldsymbol{s}_{yx}^T \boldsymbol{S}_{xx}^{-1} \bar{\boldsymbol{x}} \\ &= \; \bar{y} \; . \end{aligned} \tag{A373}$$

- The vector of *standardized coefficients* $\hat{\boldsymbol{\beta}}_1^*$ are

$$\hat{\boldsymbol{\beta}}_1^* \; = \; \frac{1}{s_y} \, \boldsymbol{D}_x \, \hat{\boldsymbol{\beta}}_1 \; = \; \boldsymbol{R}_{xx}^{-1} \, \boldsymbol{r}_{yx} \; . \tag{A374}$$

The next theorem is Theorem 10.6 in Rencher and Schaalje [100] and gives a breakdown of the factors that determine how much each variable adds to R^2 [100, Chapter 10.6, p. 263].